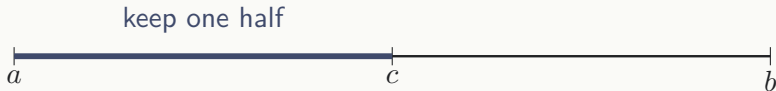


The Bisection Method

Finding a root by repeatedly halving an interval



No calculus required. We use signs, continuity, and careful bookkeeping.

Lesson goals

By the end, you should be able to:

- explain what a root means in the equation $f(x) = 0$;
- use opposite signs to trap a root inside an interval;
- carry out several bisection steps and justify each choice;
- stop when the remaining uncertainty is small enough.

Main idea

Keep an interval whose endpoints lie on opposite sides of zero, then cut that interval in half.

Roots and the equation $f(x) = 0$

A **root** is an input x that makes the function output equal to zero.

$$f(x) = 0$$

For this lesson,

$$f(x) = x^2 - 3.$$

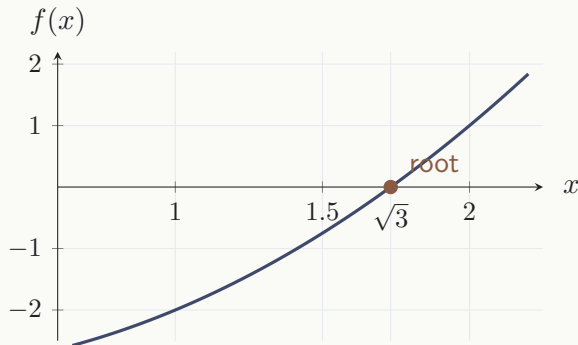
Therefore, finding a root means finding a number whose square is 3:

$$x^2 - 3 = 0 \quad \Longleftrightarrow \quad x^2 = 3.$$

Graph meaning

At a root, the graph meets the horizontal axis because its height is $f(x) = 0$.

A root is an axis crossing



$$f(1) = 1^2 - 3 = -2$$

$$f(2) = 2^2 - 3 = 1$$

At $x = 1$ the graph sits below zero.

At $x = 2$ it sits above zero.

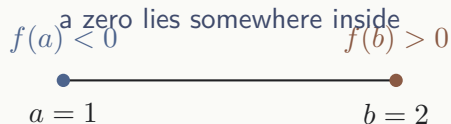
Why opposite signs matter

We begin with two inputs:

$$a = 1, \quad b = 2.$$

Their outputs have opposite signs:

$$f(a) = f(1) = -2, \quad f(b) = f(2) = +1.$$



The two endpoint values lie on different sides of zero. A continuous graph must pass through zero between them.

Continuity and the Intermediate Value Theorem

Think of drawing the graph from $x = 1$ to $x = 2$ **without lifting your pencil**.

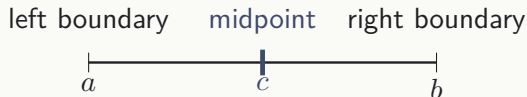
1. The graph starts below the horizontal axis because $f(1) < 0$.
2. It ends above the horizontal axis because $f(2) > 0$.
3. A continuous path cannot move from below to above without touching the axis.

Intermediate Value Theorem, in plain language

A continuous function takes every height between its starting height and ending height. Since zero lies between -2 and $+1$, the function must take the value zero.

Important: the guarantee depends on continuity over the whole interval.

The three symbols: a , b , and c



$$c = \frac{a + b}{2}$$

We evaluate $f(c)$ to learn which half still has opposite signs at its ends.

What changes after a step?

One boundary moves to c . The other boundary stays where it is. The root remains trapped between them.

Starting interval for $f(x) = x^2 - 3$

$$a = 1, \quad b = 2$$

$$f(a) = -2, \quad f(b) = +1$$

$$f(a)f(b) = (-2)(1) = -2 < 0$$

Why multiply?

A negative product means one factor is negative and the other is positive. So $f(a)f(b) < 0$ is a short way to say the endpoint values have opposite signs.

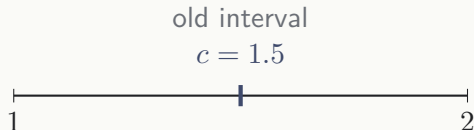
Because the function is continuous and the signs differ, we may safely begin bisection on $[1, 2]$.

Iteration 1: test the midpoint

$$c = \frac{1 + 2}{2} = 1.5$$

$$f(c) = 1.5^2 - 3 = -0.75$$

The midpoint is below zero, just like the left endpoint.



Signs:

$$f(1) < 0, \quad f(1.5) < 0, \quad f(2) > 0$$

Decision

The sign change occurs between 1.5 and 2, so the new interval is $[1.5, 2]$.

What does $f(a)f(c) < 0$ tell us?

We compare signs in the left half first.

$$f(a)f(c) < 0$$

means $f(a)$ and $f(c)$ have opposite signs. Therefore, the root lies in the left half $[a, c]$.

If the product is positive

If $f(a)f(c) > 0$, then $f(a)$ and $f(c)$ have the same sign. The left half does not preserve the sign change, so keep the right half $[c, b]$.

In iteration 1,

$$f(1)f(1.5) = (-2)(-0.75) = +1.5 > 0,$$

so we keep $[1.5, 2]$.

Iteration 2: halve the new interval

Now $a = 1.5$ and $b = 2$.

$$c = \frac{1.5 + 2}{2} = 1.75$$

$$f(1.75) = 1.75^2 - 3 = +0.0625$$

Compare the left boundary with the midpoint:

$$f(1.5)f(1.75) = (-0.75)(0.0625) < 0.$$

Decision

Opposite signs occur between 1.5 and 1.75, so the new interval is $[1.5, 1.75]$.

Iteration 3: repeat the same logic

Now $a = 1.5$ and $b = 1.75$.

$$c = \frac{1.5 + 1.75}{2} = 1.625$$

$$f(1.625) = 1.625^2 - 3 = -0.359375$$

The midpoint and a are both negative. The sign change lies between the midpoint and b .

Decision

Replace a with c . The new interval is $[1.625, 1.75]$.

Every iteration repeats the same question: *Which half still has opposite signs at its endpoints?*

Several iterations together

Step	a	b	c	$f(c)$	Kept interval
1	1.00000	2.00000	1.50000	-0.750000	[1.50000, 2.00000]
2	1.50000	2.00000	1.75000	+0.062500	[1.50000, 1.75000]
3	1.50000	1.75000	1.62500	-0.359375	[1.62500, 1.75000]
4	1.62500	1.75000	1.68750	-0.152344	[1.68750, 1.75000]
5	1.68750	1.75000	1.71875	-0.045898	[1.71875, 1.75000]
6	1.71875	1.75000	1.73438	+0.008057	[1.71875, 1.73438]

The intervals close in on $\sqrt{3} \approx 1.73205$ without needing to know that value in advance.

The uncertainty halves every time

The interval width is $b - a$. After one bisection, the new width is half as large.

$$\text{width after } n \text{ steps} = \frac{b_0 - a_0}{2^n}$$

For the starting interval $[1, 2]$, the width begins at 1:

Step	0	1	2	3	6
Width	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{64}$

If we report the midpoint, the root is at most half the current interval width away:

$$\text{maximum midpoint error} \leq \frac{b - a}{2}.$$

A simple stopping criterion

Choose a tolerance before you start. For example:

$$\varepsilon = 0.01.$$

Stop when the interval is narrow enough:

$$\frac{b-a}{2} < \varepsilon.$$

After step 6, the interval is $[1.71875, 1.734375]$.

$$\frac{b-a}{2} = \frac{0.015625}{2} = 0.0078125 < 0.01.$$

We can report its midpoint, $x \approx 1.72656$, with guaranteed error below 0.01.

Another common rule

Some problems stop when $|f(c)|$ is small. Always state which rule you use.

Bisection algorithm

Input

A continuous function f , endpoints $a < b$ with $f(a)f(b) < 0$, and a tolerance ε .

1. While $(b - a)/2 \geq \varepsilon$:
 - 1.1 Compute $c = (a + b)/2$.
 - 1.2 If $f(c) = 0$, return c .
 - 1.3 If $f(a)f(c) < 0$, set $b = c$.
 - 1.4 Otherwise, set $a = c$.
2. Return $(a + b)/2$ as the approximate root.

Invariant: throughout the loop, $f(a)$ and $f(b)$ keep opposite signs, so the root remains trapped in $[a, b]$.

Common mistakes

- **Skipping the starting sign check.** Verify continuity and $f(a)f(b) < 0$ before beginning.
- **Keeping the wrong half.** Keep the half whose endpoint function values have opposite signs.
- **Comparing a , b , and c instead of their outputs.** The sign test uses $f(a)$, $f(b)$, and $f(c)$.
- **Moving both endpoints.** At each step, replace only one endpoint with c .
- **Rounding too early.** Keep extra digits during the work, then round the final answer.
- **Forgetting the continuity condition.** A sign change alone does not guarantee a root if the function jumps.

Practice question

Use bisection to approximate a root of

$$f(x) = x^3 - x - 2$$

on the interval $[1, 2]$.

1. Show that the starting endpoint values have opposite signs.
2. Find the first three midpoints.
3. After each midpoint, state which interval you keep and why.
4. Give the interval width after three iterations.

Self-check question

At every stage, can you point to one negative endpoint value and one positive endpoint value?